DHAIRYA BHATIA

bhatiadhairya19@gmail.com · +1 (617) 503-7797 · LinkedIn · GitHub · Website

SUMMARY

ML researcher building and evaluating multimodal and video-language models in PyTorch. Focused on model evaluation and benchmarking, temporal reasoning, and large-scale experimentation on HPC.

EDUCATION

Khoury College of Computer Science, Northeastern University — Boston, MA
Master of Science in Computer Science — Sep 2025 – May 2027 (Expected)

The University of British Columbia — Kelowna, BC
Bachelor of Science in Computer Science — Sep 2020 – May 2024

RESEARCH EXPERIENCE

Augmented Cognition Lab (ACLab), Northeastern University — May 2026 – Present
Research Assistant · Advisor: Prof. Sarah Ostadabbas

- Designed and led MotionBlind, a contrastive video benchmark (100+ paired clips across 9 motion categories) for evaluating temporal and motion reasoning in Video-LLMs.
- Built multimodal VLM evaluation pipelines on HPC (SLURM) and established baselines across open-source and frontier models.
- Served as paper reviewer for the CV4Smalls Workshop at CVPR 2026.

CACTILab, Northeastern University — Mar 2026 – Present
Research Volunteer

- Designed and built a multi-stage agentic code-generation pipeline with LLM orchestration and compiler-in-the-loop verification.
- Implemented model tiering, per-turn cost observability, and multi-agent verification loops.

PUBLICATIONS

Dhairya Bhatia, Bishoy Galoaa, et al. *MotionBlind: Probing the Illusion of Motion Understanding in Video-LLMs*. arXiv:2609.09528, 2026. Under review, NeurIPS 2026 Workshop. **Co-first author.**

TEACHING

Teaching Assistant, CS3100 — Northeastern University — Jan 2026 – Jun 2026

- Supported students across labs and office hours; helped design and test labs and assignments before release.
- Led analysis of student feedback on AI-assisted programming workflows, informing curriculum design.

INDUSTRY EXPERIENCE

Oriv — Software Development Intern / AI Developer — Sep 2024 – Jul 2025

- Prototyped knowledge-graph methods for fitting graph structures onto IoT sensor signals for industrial automation POCs.
- Introduced sensor-virtualization methods enabling data simulation and reducing dependency on physical hardware.

PROJECTS

MITRE eCTF 2026 — Embedded Security Competition

- Contributed firmware implementation, code review, and bug fixes in a security-critical, hardware-constrained environment.

Project Escalion — NU Launch Labs (Co-Founder & Team Lead)

- Co-founded a Rust-based game development studio (now a Khoury-recognized club); designing a custom engine focused on high-performance rendering and real-time data pipelines.

TECHNICAL SKILLS

Languages: Python, Rust, C, Java, SQL, Bash
ML / AI: PyTorch, Multimodal Learning, VLMs, Model Evaluation & Benchmarking, Computer Vision, Distributed Training
Systems & Tools: Linux, SLURM, HPC, Docker, Git, Embedded Systems